Data Representation

OHE → One Hot Encoding

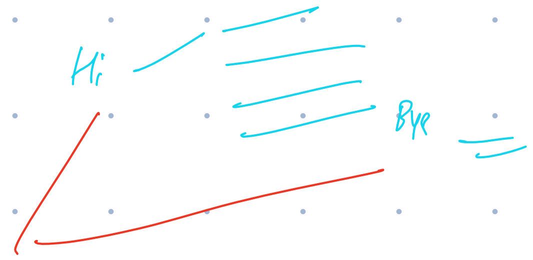
" I love Chennai , Chennai is good city "

$$
\begin{bmatrix}
1 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

10,000 words

↳ Sparse Vector

Semantic



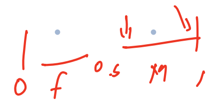
Nil

Embeddings

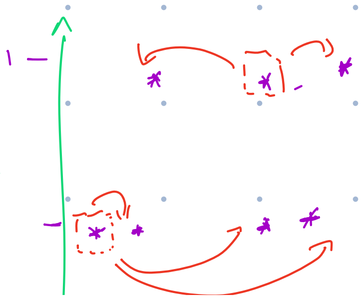
Dense Representation

King = [0.67 , 0.84 , 0.93]

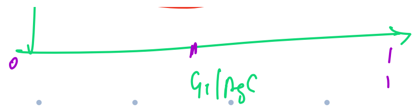
↳ feature / Dimension



	Royalty	Gender / Age
King	0.9	0.85
queen	0.9	0.35
Man	0.4	0.75
		0.30



Women	0.4	
Prince	0.9	0.65
Boy	0.35	0.65
Girl	0.4	0.15



384, 512, 1536, 3000, 12000,

↳ Embedding Model ↳ Open- Text Embedding 3 small
large
ada-002

Cohere

→ BGE
→ all miniLM

Prompt Engineering

User → intent → LLM → Doesn't

Task → What

Context → Background info

Instructions
↳ How,
3 Points
JSON

inputs → File

Zero Shot → No Guidance

Few/One Shot → Q: What is orange
A:

Chain of thought →

Critique → Ideas - Hackathon →
Develope

Role Base → Act like a [illegible]

Meta prompting →

Roles

→ User → input

→ System message → [illegible] Developer - instructions

A → Act as a — developer python — Reviewer

U → Userpersona → .net develuper

T → Task

O - output defination

M - Mode

A → Atypical use case

T → Topic whitelisting / Blacklisting

Context Engineering

→ Question → User message

System message → + → LLM

KB → LLM

Tool → LLM

Agent → LLM

DB → LLM

ChatGPT → info → GPT-5

Stateless →

LLM — large — language — Model

RAG

↳ Retrieval Augmented Generation

LLM → Knowledge cut off

→ Knowledge drift

→ public data X private data

Hallucinations

↳ Citations → Reliability

↳ Learn → inferencing

(websearch)

Tools

↓

Extend capabilities (LLM

↓

fine tuning

↳

→ LLM 1B

↓

finetuned

→ RAG

Query → LLM

⇓

Query → Retrieval

⇓

LLM

↳ Answer

Data Ingestion

200-800

Document loaders ⇒

0 500

C1	C2	C3	C4	C5

400

Chunking ⇒

[—]

Embedding model → → Database

1536

↓

E1	E2	E3

1 milion

Vector

E1-C1

E2-C2

E3-C3

Chroma

Pinecone

Weaviate

Milvus

fixed size chunking

" " " overlap

Sentence

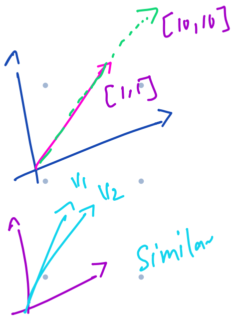
Paragraph

Context aware

Semantic Chunking

Recursive text splitter

(500 Token



Retrieval

Top k=3

Question → Embedding Model → Q_V

[ ]

Cosine Similarity

VDB

E1 E2
E3
E4 E5

E1 = 0.9

E2 = 0.6

E3 = 0.8

E4 = 0.75

$$= \frac{A \cdot B}{|A|\,|B|} = 0.001$$

1 = Similar

0 = Not Simi

-1 = opposite

Q = T C1

T C3

T C4

T

Generation

Prompt = Inst + Q + C1 + C3 + C4

LLM